Name: Vansh Sharma **Batch: T22**
Roll No.: 2401109 **Subject: ASD&D**

EXPERIMENT - 1

Aim: Study and Implementation of Basic Git and GitHub Operations.

This Experiment maps to LO2.

LO2: To understand and perform the basic version control operations using Git and GitHub, including repository creation, cloning, file tracking, committing, pushing, pulling, branching, merging, and remote repository management.

Theory:
Git is a distributed version control system that helps developers track changes in source code and files. It enables efficient project management through repositories, commits, branches, and remote repositories, making collaboration and version management easier. In this experiment, various Git commands are performed to understand the basic operations of version control.

Commands:
git init : Initializes a new Git repository in the current working directory.

git clone <repository-url> : Creates a local copy of an existing remote GitHub repository.

git status : Displays the current state of the repository, including staged, modified, and untracked files.

git add <file> : Adds a specific file to the staging area before committing.

git add . : Stages all newly created and modified files in the current directory.

git commit -m "message" : Records the staged changes in the local repository with a descriptive commit message.

git log : Shows the complete history of commits made in the repository.

git log --oneline : Displays a concise, one-line summary of each commit.

git diff : Shows the differences between the working directory and the most recent commit.

git branch : Lists all available local branches and indicates the active branch.

git branch <branch-name> : Creates a new branch without switching to it.

git checkout <branch-name> : Switches from the current branch to the specified branch.

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git checkout -b <branch-name> : Creates a new branch and immediately switches to it.

git switch <branch-name> : Switches to an existing branch using the modern Git command.

git merge <branch-name> : Merges the specified branch into the currently active branch.

git remote -v : Displays the remote repositories linked to the local repository.

git remote add origin <repository-url> : Connects the local repository to a remote GitHub repository.

git push origin <branch-name> : Uploads local commits from the specified branch to the remote repository.

git pull origin <branch-name> : Fetches and merges the latest changes from the remote repository into the current branch.

git fetch : Downloads updates from the remote repository without merging them into the current branch.

git reset --hard HEAD : Removes all uncommitted changes and restores the working directory to the latest commit.

git stash : Temporarily saves uncommitted changes without creating a commit.

git stash pop : Restores the most recently stashed changes and removes them from the stash list.

git rm <file> : Deletes a file from both the working directory and the Git repository.

git config --global user.name "Your Name" : Sets the global username that Git uses for commits.

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```
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (main) $ /workspaces/ASD-D_Lab/Exp1
bash: /workspaces/ASD-D_Lab/Exp1: No such file or directory
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (main) $ /workspaces/ASD-D_Lab/Exp1
Hello World!!@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (main) $ git add .
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (main) $ git commit -m "Added Exp1"
[main 625eabc] Added Exp1
 3 files changed, 35 insertions(+)
 create mode 100644 .vscode/tasks.json
 create mode 100755 Exp1
 create mode 100644 Exp1.cpp
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (main) $ git push
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 13.39 KiB | 3.35 MiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/HammadShaikh-051/ASD-D_Lab
   682ebeb..625eabc  main -> main
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (main) $ git switch -c hammad
Switched to a new branch 'hammad'
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (hammad) $ git add .
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (hammad) $ git commit -m "Commiting from new Bran
ch"
[hammad 8045400] Commiting from new Branch
 1 file changed, 2 insertions(+)
```

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```
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (hammad) $ git push -u origin hammad
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 438 bytes | 438.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'hammad' on GitHub by visiting:
remote:      https://github.com/HammadShaikh-051/ASD-D_Lab/pull/new/hammad
remote:
To https://github.com/HammadShaikh-051/ASD-D_Lab
 * [new branch]      hammad -> hammad
branch 'hammad' set up to track 'origin/hammad'.
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (hammad) $ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (main) $ git merge hammad
Updating 625eabc..8045400
Fast-forward
 Exp1.cpp | 2 ++
 1 file changed, 2 insertions(+)
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (main) $ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/HammadShaikh-051/ASD-D_Lab
   625eabc..8045400  main -> main
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (main) $ git switch hammad
Switched to branch 'hammad'
Your branch is up to date with 'origin/hammad'.
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (hammad) $ git branch
* hammad
  main
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (hammad) $ git switch main
Switched to branch 'main'
```

```
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (main) $ git merge hammad
Fast-forward
 Exp1.cpp | 2 ++
 1 file changed, 2 insertions(+)
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (main) $ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/HammadShaikh-051/ASD-D_Lab
   625eabc..8045400  main -> main
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (main) $ git switch hammad
Switched to branch 'hammad'
Your branch is up to date with 'origin/hammad'.
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (hammad) $ git branch
* hammad
  main
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (hammad) $ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (main) $ git branch
  hammad
* main
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (main) $ git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 4 (delta 2), reused 2 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 1.03 KiB | 262.00 KiB/s, done.
From https://github.com/HammadShaikh-051/ASD-D_Lab
   8045400..b24af87  main       -> origin/main
 * [new branch]      vansh      -> origin/vansh
Updating 8045400..b24af87
Fast-forward
 hello.py | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 hello.py
@HammadShaikh-051 ➜/workspaces/ASD-D_Lab (main) $
```

CONCLUSION:

The experiment demonstrated the use of essential Git version control commands for repository management, tracking changes, branch handling, and collaboration with GitHub. It provided practical knowledge of managing and maintaining software projects efficiently using Git.